

Movie Economics & Ratings Analytics

Top 200 Highest-Grossing Films - A TMDB x OMDB Data Analysis Pipeline

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Introduction & Objective

This report presents an end-to-end analytics pipeline built on live movie data from TMDB and OMDB, combining box-office economics with critical and audience reception for the top 200 highest-grossing films.

Objectives

- Understand the distribution of budgets, revenue, ROI, and ratings across 200 films
- Analyze whether bigger budgets lead to better reception
- Measure agreement between IMDb, Rotten Tomatoes, and Metascore
- Identify genre-level and studio-level patterns
- Segment films by budget bracket with data-backed recommendations

Tools & Environment

Tool / Library	Purpose
Python 3	Core programming language
Requests + python-dotenv	Live TMDB & OMDB API connectivity
Pandas	Data manipulation & cleaning
Numpy	Numerical aggregation & correlation
Matplotlib / Seaborn	Visualization
VS Code	Development environment

Dataset Overview

200

Movies Pulled (TMDB)

200

Movies Cleaned & Verified

100%

Ratings Matched (OMDB)

25

Master Table Columns

Data pulled live via TMDB's /discover/movie endpoint (revenue-sorted, 10 pages), with a second call to /movie/{id} for budget, revenue, and imdb_id. Rating data pulled from OMDB per imdb_id, then joined into a single master table.

Coverage

IMDb rating matched	198 / 200 (99.0%)
Usable budget reported	199 / 200 (99.5%)
Usable revenue reported	200 / 200 (100.0%)

Key columns include budget/revenue (with zero-flagging), ROI, genres, runtime, IMDb rating, Rotten Tomatoes score, Metascore, and production companies.

Data Cleaning - Steps & Decisions

Cleaning Decision	Reason
Zero budgets/revenue	Flagged & nulled, not treated as real zeros
Duplicate rows	Dropped on tmdb_id and imdb_id
ROI calculation	Computed only where both budget & revenue reported
“N/A” strings (OMDB)	Converted to proper nulls before numeric conversion
Formatted numeric fields	Commas, % and \$ signs stripped and converted to numeric
Runtime = 0	Treated as missing, set to null

Post-cleaning verification confirmed: no duplicate IDs, release years within 1900-2027, ratings within valid ranges, and zero leftover “N/A” strings. Join coverage: 100% of cleaned OMDB rows matched an existing movie record.

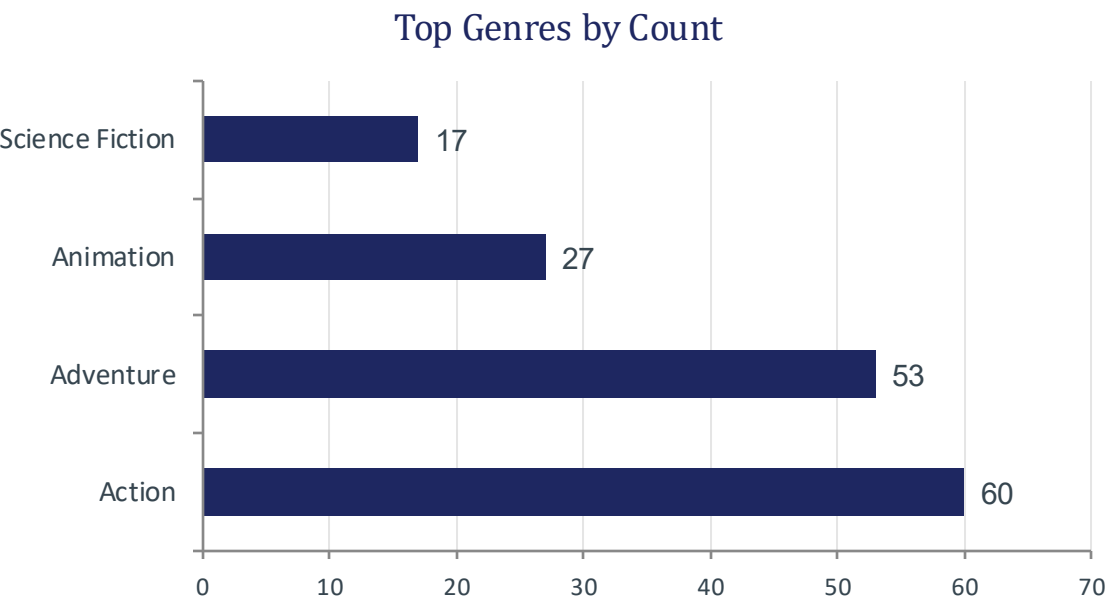
Exploratory Data Analysis - Key Metrics



Metric	IMDb (/10)	RT (%)	Budget	Revenue	ROI
Min	4.8	18	\$0.2M	\$619.0M	0.52x
25th %ile	6.6	60.75	\$111.5M	\$719.1M	3.20x
Median	7.3	79	\$160.0M	\$852.7M	4.53x
Mean	7.20	74.08	\$167.4M	\$954.9M	23.54x
75th %ile	7.8	91	\$200.0M	\$1,040.1M	7.49x
Max	9.1	99	\$658.8M	\$2,923.7M	3,349x

ROI is heavily right-skewed - the mean (23.54x) is pulled far above the median (4.53x) by a handful of extreme low-budget breakout hits. The median is the more representative figure for a typical top-200 film.

Genre & Certification Snapshot

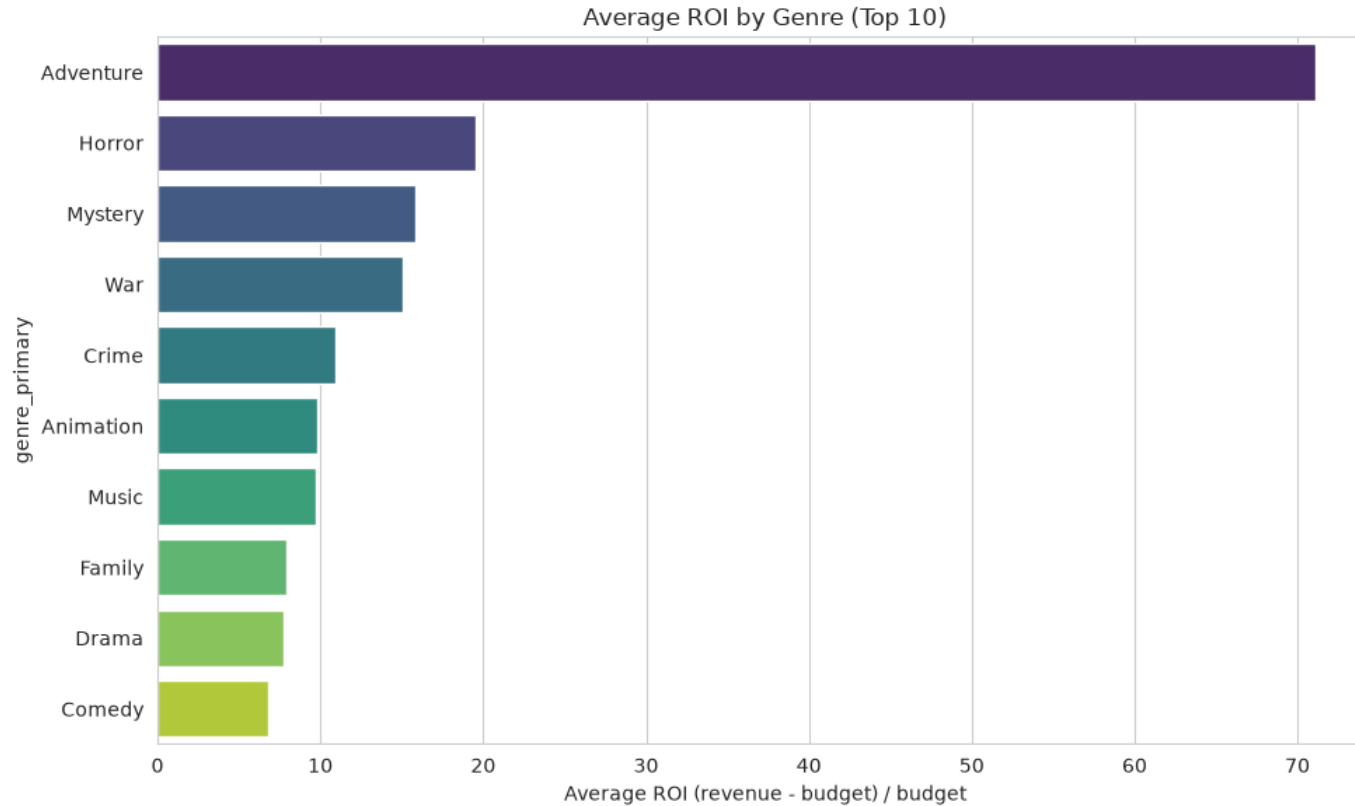


MPAA Rating

PG-13	116 (58%)
PG	56 (28%)
R	13 (6.5%)
G	6 (3%)
Not Rated	5 (2.5%)

Original language: English 94.5%, Chinese 5.0%, Japanese 0.5%.
Runtime: median 131 min, mean 130.3 min, range 86-201 min.

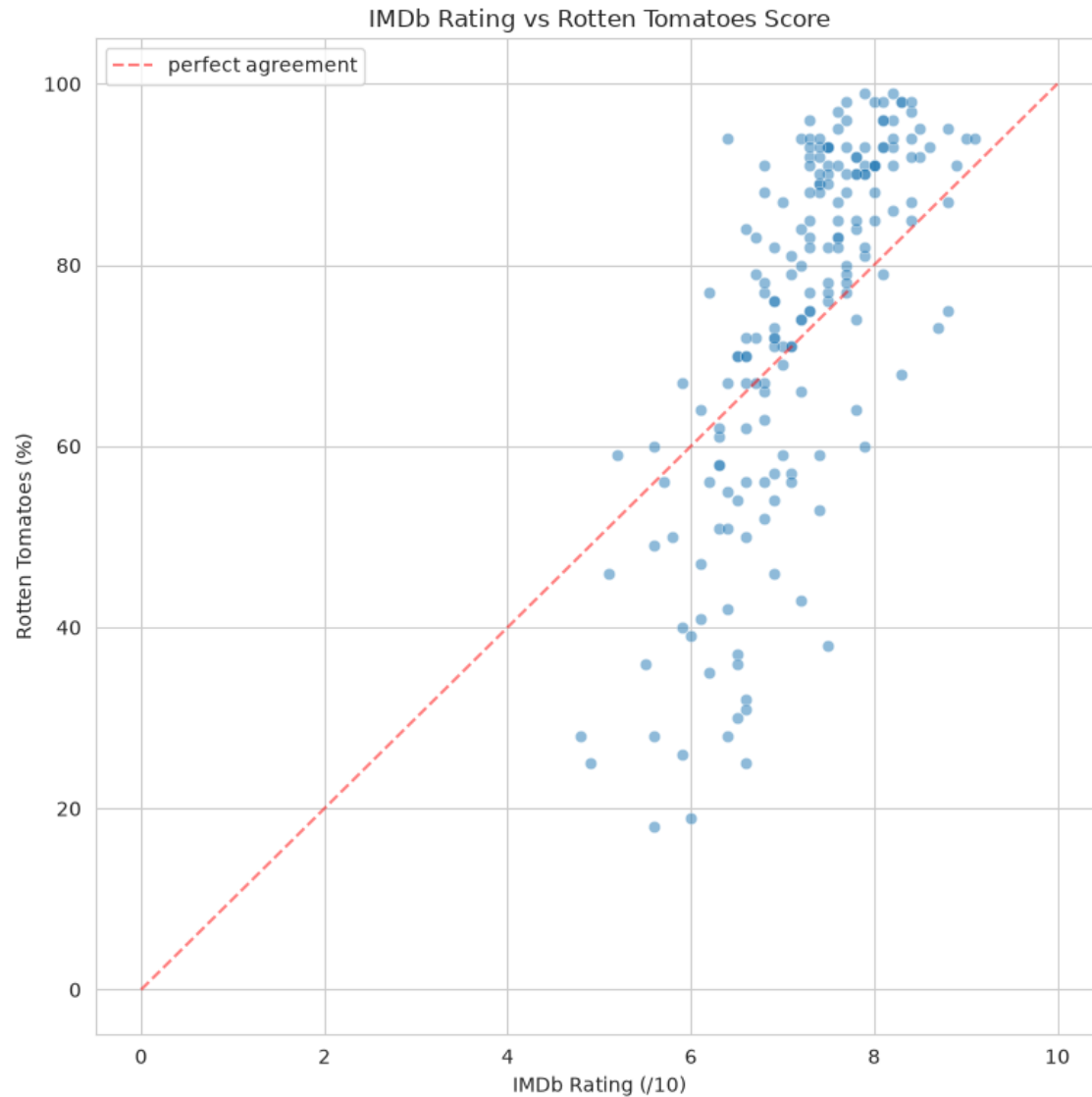
Average ROI by Genre



Key Takeaways

- Outlier-driven leader: Adventure averages ~71x ROI, pulled up by a handful of extreme low-budget outliers.
- Horror, Mystery, and War form a distant second tier (15-20x), all modest-budget genres.
- High-volume genres (Comedy, Drama, Family) return the least per dollar (7-8x), though still profitable.

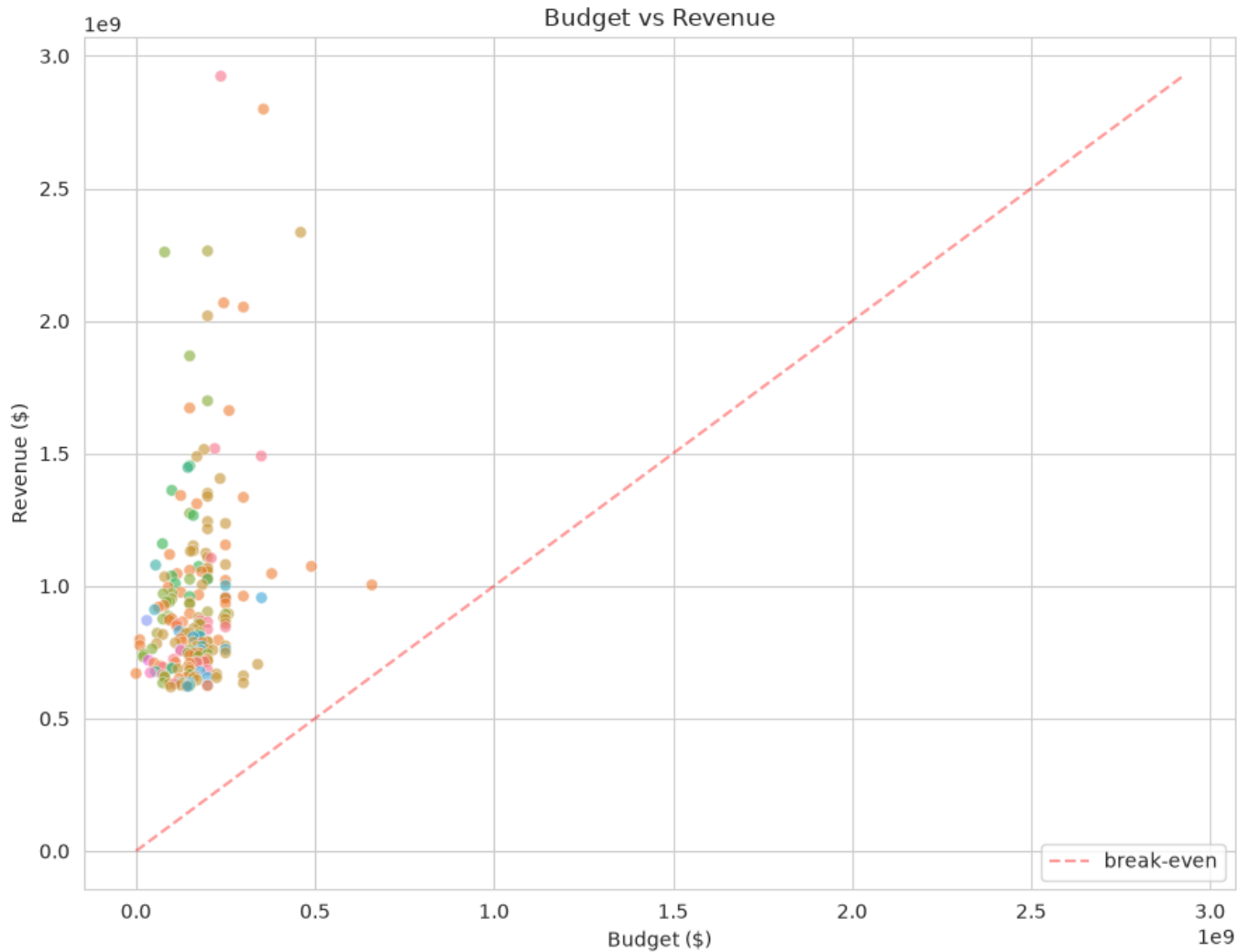
IMDb Rating vs Rotten Tomatoes Score



Key Takeaways

- Broad agreement: across 192 films, IMDb and RT correlate at 0.75, a fairly strong relationship.
- A mild critic-friendly skew: RT scores tend to run slightly more generous than equivalent IMDb ratings.
- Notable divergence: Transformers: Revenge of the Fallen is the extreme case - IMDb 6.0/10 vs RT 19%.

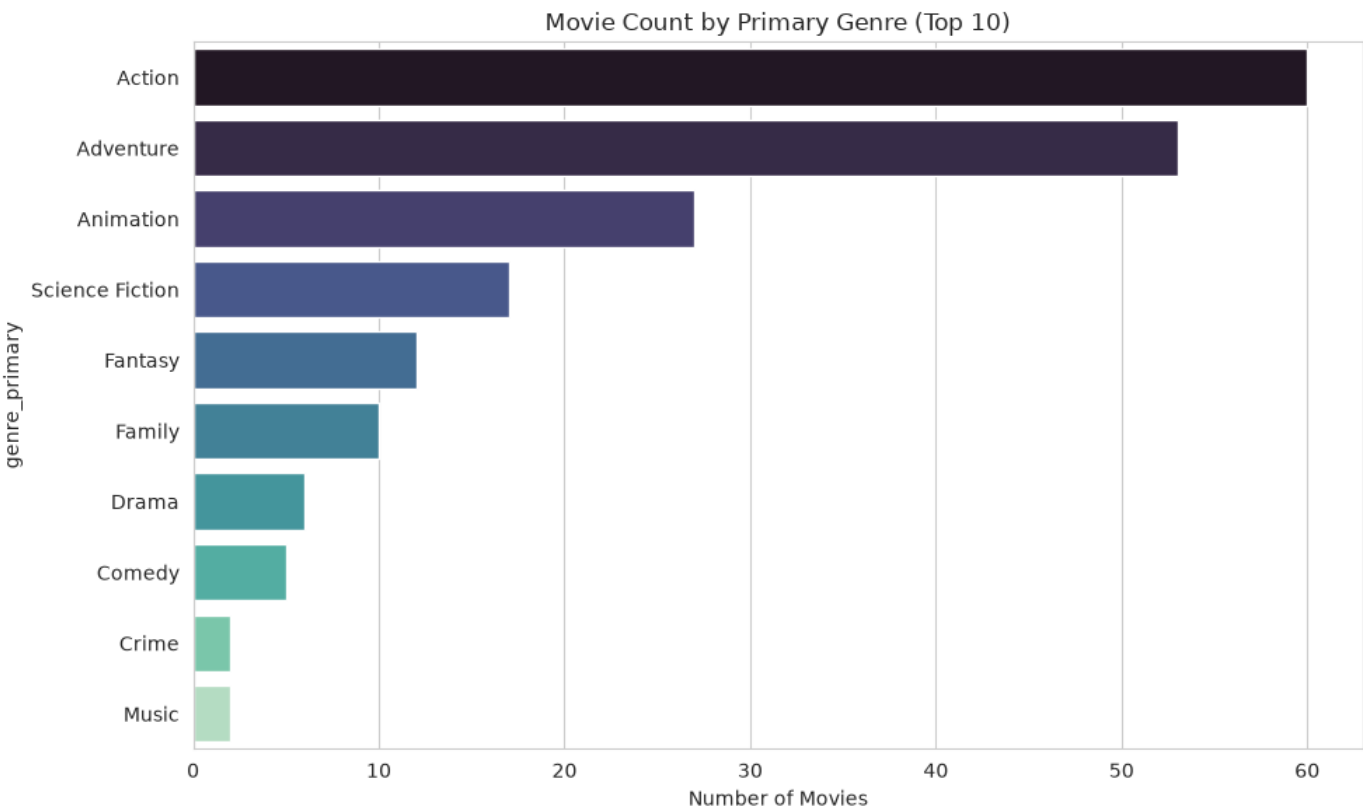
Budget vs Revenue



Key Takeaways

- Selection effect: every film sits above break-even by construction - this is a highest-grossing list, not evidence all films are profitable.
- Mid-high budgets punch above their weight: several \$2B+ revenue outcomes come from \$200-350M budgets, not the largest.
- Budget is a weak revenue predictor: correlation is a moderate 0.31.

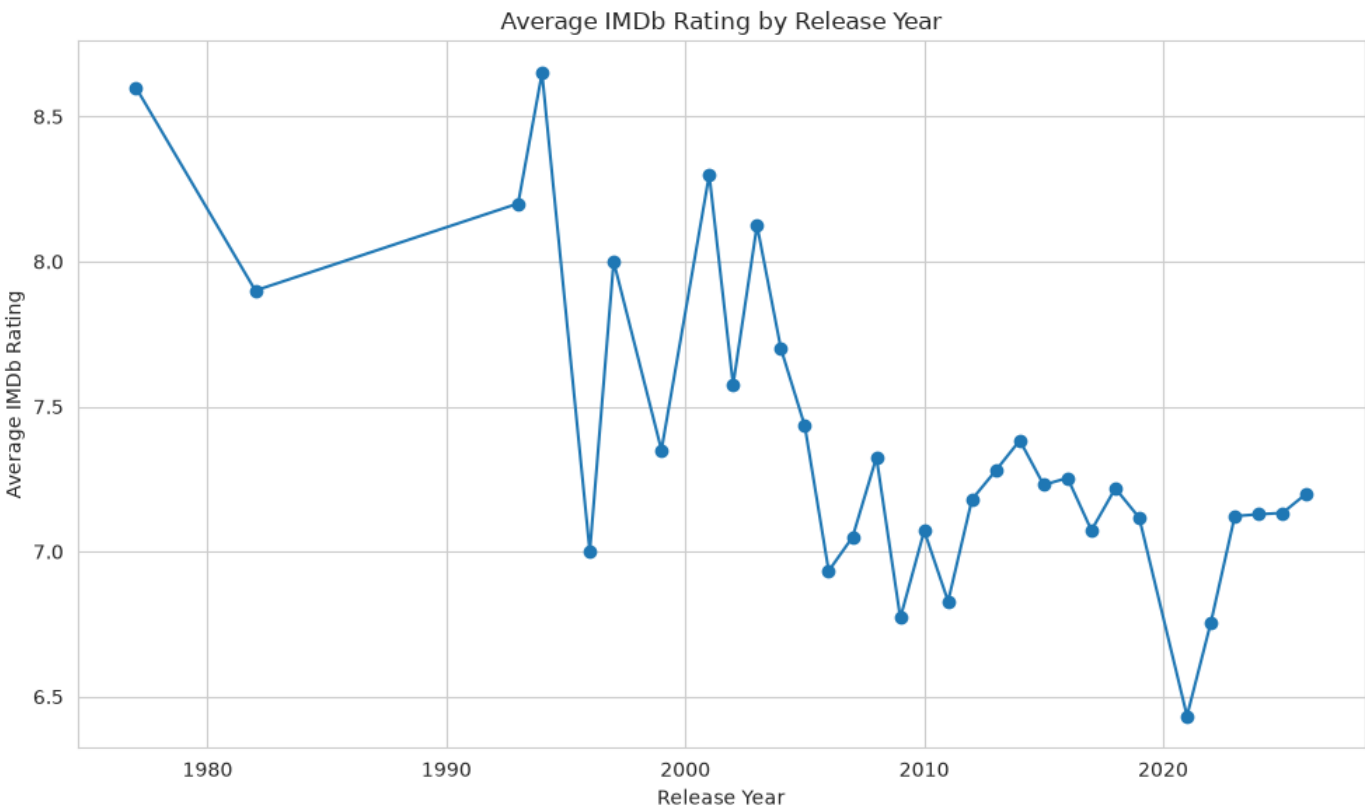
Movie Count by Genre



Key Takeaways

- Action (60) and Adventure (53) together make up 56.5% of the entire top-200 list.
- Animation (27) is a strong third tier, and carries the highest average IMDb rating among high-volume genres.
- Prestige and niche genres (Drama, Comedy, Crime, Horror) are rare in the highest-grossing bracket.

Average IMDb Rating by Release Year



Key Takeaways

- Clear long-term decline: average IMDb rating fell from 8.60 (1977) to 7.20 (2026).
- Early years are thin samples: 1977 and 1994 are represented by only one or two iconic titles.
- Recent years have stabilized: from ~2010 onward, the yearly average settles into a narrow 6.4-7.4 band.

Genre & Studio Analysis

Average Rating by Genre

Genre	Films	Avg IMDb	Avg RT
Animation	27	7.47	82.4%
Adventure	53	7.25	72.1%
Science Fiction	17	7.24	72.0%
Drama	6	7.20	76.3%
Action	60	7.17	75.7%
Comedy	5	7.06	75.8%
Fantasy	12	6.88	63.8%
Family	10	6.83	72.4%

Top Studios by Volume

Studio	Films
Marvel Studios	24
Warner Bros. Pictures	15
Walt Disney Pictures	14
Pixar	13
Lucasfilm Ltd.	8
Universal Pictures	8
DreamWorks Animation	8

Marvel Studios, Warner Bros., Disney, and Pixar together account for 66 of 200 films (33.0%), reflecting how concentrated franchise-driven filmmaking is at the top of the box office.

Budget Bracket Analysis

Bracket	Range	Films	Share	Avg ROI	Avg IMDb
Lean	Under \$100M	38	19.1%	104.2x	7.42
Mid-Budget	\$100M-\$200M	93	46.7%	5.15x	7.21
Tentpole	Above \$200M	68	34.2%	3.59x	7.09

Observations

- The efficiency sweet spot: even discounting the Lean bracket's outlier-driven average, the Mid-Budget tier offers the best combination of scale, consistency, and return.
- Bigger budgets trend toward slightly lower ratings: average IMDb rating declines steadily as budget bracket increases (7.42 to 7.21 to 7.09).
- Tentpoles still work, just less efficiently: they deliver strong absolute revenue but remain the smallest-ROI-multiple bracket.

Real-World Problems & Solutions

Problem 1: Budget Does Not Buy Better Reception

Data: Budget correlates with IMDb rating at -0.13, Rotten Tomatoes at -0.20, Metascore at -0.15 - all weak and slightly negative.

Solution: Marginal spend beyond a point should not be justified by expected reception gains. Story, casting, and execution matter more than budget scale.

Problem 2: Diminishing ROI Efficiency at the Top

Data: Average ROI falls steadily from Lean to Tentpole, while the Mid-Budget tier captures nearly half the list with a strong, repeatable 5.15x return.

Solution: A portfolio balancing fewer tentpole bets with a larger base of mid-budget films is likely more capital-efficient.

Problem 3: Single-Source Rating Risk

Data: IMDb and Rotten Tomatoes correlate at 0.75 overall, but individual titles diverge sharply (e.g. Transformers: Revenge of the Fallen: 6.0/10 IMDb vs 19% RT).

Solution: Blend audience (IMDb) and critic (RT, Metascore) signals rather than relying on a single rating source for key decisions.

Strategic Recommendations

- 1 Prioritize the Mid-Budget Tier**
Best combination of volume (46.7%), consistency, and average ROI (5.15x).
- 2 Do Not Equate Budget With Quality**
Budget correlates weakly, and slightly negatively, with all three rating measures.
- 3 Track Multiple Rating Sources**
IMDb, Rotten Tomatoes, and Metascore only pairwise-correlate 0.66-0.89; use a blended score.
- 4 Lean Into Animation's Rating Advantage**
Highest average IMDb rating (7.47) among high-volume genres, and comparatively under-leveraged.
- 5 Recognize Studio Concentration**
Marvel, Warner Bros., Disney, and Pixar already account for a third of the top-200 list.

Technical Challenges & Problem Solving

Dual-API Join Complexity

TMDB's list endpoint lacks budget/revenue/imdb_id. Solved with a second per-movie API call in fetch.py.

Rate-Limited Free-Tier APIs

Both TMDB and OMDb free tiers are rate-limited. Small delays added between calls to avoid throttling.

Inconsistent Missing-Value Encoding

OMDb returns the string "N/A" instead of a null. An explicit na_like check handles this before numeric conversion.

False Zeros in Budget & Revenue

TMDB returns 0 for unreported fields, not real zeros. Booleans flag reported values before nulling zeros.

Post-Cleaning Verification Layer

verification.py independently re-checks each table before the join, catching upstream cleaning bugs early.

Conclusion

This analysis joins live TMDB budget/revenue data with live OMDb rating data into a single, verified master dataset, transforming two separate API responses into actionable film-economics intelligence. Within this highest-grossing slate, budget scale is a weak predictor of both revenue outcome and critical reception, while the mid-budget tier and a small number of dominant studios and genres, Animation in particular, carry the strongest quality signals.

Data Disclaimer

Data pulled from third-party APIs is never perfectly complete. This pipeline preserves that reality rather than hiding it: unreported budget/revenue is explicitly flagged, and 2 of 200 films carry no matched IMDb rating. Because this dataset is TMDB's top 200 films sorted by revenue, all insights describe patterns within an already highly successful slate, not the broader film market. Correlations describe association, not causation.